6. **(a)** Factorise completely $(x+8)^3 - x^3$. [2]

(b) The length of the side of cube A is 8 cm longer than the length of the side of cube B.
The difference between the volumes of the two cubes is 1304 cm^3.
Using your result from **part (a)**, find the length of the side of cube A. [3]

7. The path of a water jet can be modelled by the quadratic function $h = -2x^2 + 10x + 7$, where x m is the horizontal distance it travels and h m is the height of the water above horizontal ground.

(a) State the initial height of the water. [1]

(b) Express h in the form $a + b(x+c)^2$ where a, b and c are constants to be determined. [3]

(c) Hence, explain whether the water could reach a height that is 20 m above the ground. [1]

(d) Find the maximum horizontal distance, in metres, travelled by the path of the water jet. [2]

(e) John sketched a quadratic graph using the equation $h = -2x^2 + 10x + 7$.
He claimed that if the maximum horizontal distance travelled by the path of water is p metres, then the line of symmetry of the graph can be represented by $x = \frac{p}{2}$.
Briefly explain his mistake and state the correct equation of the line of symmetry. [2]